

Ranadeep Biswas, Ph.D.

CS Researcher & Engineer

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EXPERIENCE

Informal Systems *July 2021 – Present*

Emerald Developer

Sept 2025 – Present

Modular framework for institutional blockchains

Malachite (Tendermint), Commonware (Simplex), Ethereum execution.

Projects: Onemoney Network, Initiativ.

ibc-starknet Developer

July 2024 – Aug 2025

IBC protocol implementation in Cairo for Starknet

Starknet stack, Cairo, zero-knowledge systems, production deployment.

ibc-rs Developer

July 2023 – June 2024

Core IBC implementation in Rust for Cosmos ecosystem

Trustless interoperability stack, protocol engineering, open-source project management.

Blockchain Auditor

Jan 2023 – June 2023

Security audits for Cosmos SDK, Tendermint, CosmWasm protocols

DeFi attack vectors, protocol design patterns, formal audit methodology.

Projects: Gravity Bridge, IBC Spec, Osmosis, Mars Protocol, Namada, Axelar.

Testing Engineer

July 2021 – Dec 2022

Specifications and formal verification for blockchain protocols

TLA+, Apalache model checker, model-based testing, test suite design.

EDUCATION

Ph.D. in Computer Science

Nov 2017 – March 2021

University of Paris (IRIF), France

Thesis: Automated Formal Testing of Storage Systems and Applications

Master of Science in Computer Science

Aug 2015 – April 2017

Chennai Mathematical Institute, India

Software verification using CBMC

Bachelor of Science in Mathematics and Computer Science

Aug 2012 – April 2015

Chennai Mathematical Institute, India

INTERESTS

Blockchain interoperability · Consensus protocols · Formal verification · Automated testing ·
DeFi security · Verifiable AI · Open-source contributions · Rust · Linux